

Shyamal Dharía

[\[GitHub\]](#) [\[Google Scholar\]](#) [\[LinkedIn\]](#) [\[Email\]](#) [\[Academic Website\]](#)

EDUCATION

University of Winnipeg

Master of Science (MSc) in Applied Computer Science

Winnipeg, Canada

2022 – 2024

- GPA: 4.4/4.5—Graduated with Highest Distinction

Gujarat Technological University

Bachelor of Engineering in Electronics & Communication

Surat, India

2017 – 2021

- GPA: 7.9/10

RESEARCH EXPERIENCE

University of Winnipeg

Senior Research Assistant, Applied Computer Science and Psychology

Winnipeg, Canada

2025 – Present

- Conduct machine learning and statistical analysis on multimodal neuroimaging data (EEG, fMRI, sMRI) to study biomarkers of Alzheimer's disease genetic risk (APOE- ϵ 4, PICALM rs3851179), ASMR, and other neurological conditions.
- Develop low-latency EEG classifiers using differentiable logic gate networks for edge devices, receiving the Best Paper Award at the Canadian AI Conference 2026.
- Build self-supervised Joint-Embedding Predictive Architecture (JEPA) models for zero-shot harmonization of multi-site resting-state fMRI.
- Collaborate with industry partner InteraXon (Muse) and with Carleton University on joint research projects.

Red River College Polytechnic

Mitacs Business Strategy Intern, Labs4 TRL-UP Program

Winnipeg, Canada

Sept 2026 – Present

- Developing NeuroScreen, an EEG and AI decision-support prototype for cognitive impairment screening, deployed on an NVIDIA Jetson Orin Nano with OpenBCI hardware.
- Leading Build-Measure-Learn development sprints toward a minimum viable product, with guidance from an industry mentor and a technical advisor.

Misericordia Health Center

Mitacs Accelerate Intern

Winnipeg, Canada

May 2024 – Dec 2024

- Developed an EEG data collection pipeline using PsychoPy with custom experiment integration, enabling two successful EEG-based emotion regulation studies with 60+ participants using wireless OpenBCI devices.
- Designed and trained machine learning models for emotion recognition and intensity prediction, leading to two conference publications.

PUBLICATIONS

Best Paper Award:

- ***Differentiable Logic Gate Networks for Low-Latency EEG Classification on Edge Devices***
S. Y. Dharía; S. D. Smith; C. E. Valderrama (Oral Presentation, CAI 2026; 20% acceptance rate)
Canadian AI Conference (CAI), 2026

Pre-prints or Under Review:

- ***Zero-Shot Harmonization of Multi-Site Resting-State fMRI with Joint-Embedding Predictive Architectures***
S. Y. Dharía; S. D. Smith; C. E. Valderrama
Under Review (Journal)
- ***Cross-Cohort Transfer of Clinical Alzheimer's EEG Signatures to Pre-Clinical APOE- ϵ 4 and PICALM rs3851179 Risk Carriers Using Differentiable Logic Gate Networks***
S. Y. Dharía; C. E. Valderrama; S. D. Smith
Under Review (Journal)
- ***Uncertainty-Guided Gate Geometry for Low-Complexity Differentiable Logic Networks***
M. Y. M. Yassin; S. Y. Dharía; M. Taha; C. E. Valderrama; S. D. Smith
Under Review (IEEE ICASSP 2027)

Journal Publications:

- **Exploring the Interaction of APOE- ϵ 4 and PICALM rs3851179 With Dynamic Functional Connectivity in Healthy Middle-Aged Adults at Risk for Alzheimer's Disease**
S. Y. Dharia; C. E. Valderrama; Q. Liu; S. D. Smith
Journal of Neural Engineering (IOP), 2026
<https://doi.org/10.1088/1741-2552/ae4926>
- **A Novel Approach for the Early Identification of Genetic Risk Factors for Alzheimer's Disease Using EEG and Psychometric Data**
S. Y. Dharia; Q. Liu; S. D. Smith; C. E. Valderrama
IEEE Journal of Biomedical and Health Informatics (JBHI), 2025
<https://doi.org/10.1109/JBHI.2025.3639217>
- **Fractal Dimension of Resting-State EEG as a Biomarker for Autonomous Sensory Meridian Response (ASMR) — (Oral Presentation, BHI 2025; 9.2% acceptance rate)**
S. Y. Dharia; C. E. Valderrama; Q. Liu; B. K. Fredborg; A. S. Desroches; S. D. Smith
IEEE Journal of Biomedical and Health Informatics (JBHI), 2025
<https://doi.org/10.1109/JBHI.2025.3612301>
- **Dual-Transformer Cross-Attention Framework for Alzheimer's Disease Detection via dPTE-Guided EEG Channel Selection and Multi-Modal Integration**
S. Y. Dharia; Q. Liu; S. D. Smith; C. E. Valderrama
Biomedical Signal Processing and Control, 2025
<https://doi.org/10.1016/j.bspc.2025.108390>
- **Bipartite Graph Adversarial Network for Subject-Independent Emotion Recognition**
M. Niaki; S. Y. Dharia; Y. Chen; C. E. Valderrama
IEEE Journal of Biomedical and Health Informatics (JBHI), 2025
<https://doi.org/10.1109/JBHI.2025.3570187>

Conference Publications:

- **CNN-Based Chagas Disease Detection with 12-lead ECG**
S. Y. Dharia; M. Hojjati; S. Rahman; M. M. T. Nur; C. E. Valderrama
Computing in Cardiology Conference (CinC) 2025 — PhysioNet Challenge
[\[Link\]](#)
- **Using Machine Learning to Model EEG-Derived Brain Activity During Emotion Regulation**
M. Hojjati; S. Y. Dharia; S. G. Camorlinga; S. D. Smith; A. S. Desroches; B. D. Brenneman
Proceedings of the 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), Copenhagen, Denmark, 2025
<https://doi.org/10.1109/EMBC58623.2025.11253722>
- **Leveraging Machine Learning and Threshold-Free Cluster Enhancement to Unravel Perception of Emotion and Implied Movement**
S. Y. Dharia; M. Hojjati; S. G. Camorlinga; S. D. Smith; A. S. Desroches
IEEE EMBS International Conference on Biomedical and Health Informatics (BHI), Houston, TX, USA, 2024
<https://doi.org/10.1109/BHI62660.2024.10913664>
- **Dataset-Independent EEG Channel Selection for Emotion Recognition**
S. Y. Dharia; S. G. Camorlinga; C. E. Valderrama; M. Hojjati
Proceedings of the 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), Orlando, FL, USA, 2024
<https://doi.org/10.1109/EMBC53108.2024.10782444>
- **Multimodal Deep Learning Model for Subject-Independent EEG-Based Emotion Recognition**
S. Y. Dharia; C. E. Valderrama; S. G. Camorlinga
IEEE Canadian Conference on Electrical and Computer Engineering (CCECE), Regina, SK, Canada, 2023
<https://doi.org/10.1109/CCECE58730.2023.10289007>

GRANT WRITING EXPERIENCE

Contributed to \$112,500 in awarded funding across four grants.

Mitacs Accelerate (2023) | \$60,000 (Awarded)

- Co-drafted the initial research-industry proposal in collaboration with PIs for a successful partnership between the University of Winnipeg and Misericordia Health Center.

NSERC Alliance (2026) | \$30,000 (Awarded)

- Contributed to the drafting of a collaborative grant proposal involving the University of Winnipeg and InteraXon.

UWinnipeg Major Research Grant (2026) | \$7,500 (Awarded)

- Assisted the PI in preparing the proposal to support direct research costs and laboratory operations.

Labs4 TRL-UP, Mitacs Business Strategy Internship (2026) | \$15,000 (Awarded)

- Led the proposal for NeuroScreen, an EEG and AI decision-support tool for cognitive impairment screening, securing stipend and project funding through the Labs4 TRL-UP commercialization program at Red River College Polytechnic.

AWARDS

- 2026 - Best Paper Award, Canadian AI Conference (CAI 2026)
- 2025 - Graduate Student of Highest Distinction
- 2024 - EMBC NextGen Scholar

RESEARCH COMPETITIONS

NeurIPS 2026 EEG/EMG Foundation Challenge — Team Lead, NeuroWinn 2026 – Present

- Leading team NeuroWinn in the NeurIPS 2026 competition on foundation models for EEG and EMG signals.

George B. Moody PhysioNet Challenge 2026 — Team Lead, PhysioWinn 2026

- Predicted future cognitive impairment from overnight polysomnography (sleep EEG), tying for 9th of 100 teams in the official validation phase (test-set results pending).

PhysioNet/Computing in Cardiology Challenge 2025 — Team Lead, PhysioWinn 2025

- Led a team detecting Chagas disease from 12-lead ECG using CNN-based morphological features and lead-wise learning.

TEACHING EXPERIENCE

- TA—GACS-7203 Pattern Recognition (Fall 2024 and 2023—Dr. Camilo E. Valderrama)
- TA—GACS-7206 Advanced Machine Learning (Winter 2024—Dr. Sheela Ramanna)

ACADEMIC SERVICE

- Associate Editor—EMBC 2026
- Peer Reviewer—Scientific Reports (2026)
- Peer Reviewer—International Journal of Human-Computer Interaction (2026)
- Peer Reviewer—Public Library of Science (PLOS) Aging and Health (2026)
- Peer Reviewer—IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 26)
- Peer Reviewer—IEEE EMBS International Conference on Biomedical and Health Informatics (BHI 26 and 25)

TECHNICAL SKILLS

Programming Languages: Python (PyTorch, scikit-learn, NumPy, pandas) and C++ (CUDA)

Signal Processing & Neuroimaging: MNE-Python, Nilearn, SciPy, fMRIPrep

Software & Development: Linux/Windows, Git, Docker

Hardware & Experiments: OpenBCI, NVIDIA Jetson Orin Nano, PsychoPy